

ACE TAXI DISPATCH

Unified Production Dashboard Plan

A production plan for combining WhatsApp ride ingestion and OTS supplier portal automation into one central dispatch dashboard with driver recommendations, linked ride intelligence, auto cleanup, calendar monitoring, and future auto bidding.

1. Executive Summary

The goal is to create one professional dispatch control panel where rides from WhatsApp groups and the OTS supplier portal are normalized into a single operational workflow. Operators will be able to see live rides, review incomplete jobs, approve driver recommendations, monitor schedules, track linked ride opportunities, and view calendar status from one dashboard.

WhatsApp Groups + OTS Supplier Portal

- > Unified Ride Pipeline
- > Central Database
- > Operations Dashboard
- > Recommendation + Assignment
- > Driver/Vehicle Schedules
- > Calendar + Reports

2. Current Systems To Combine

WhatsApp Dispatch System

Extracts rides from WhatsApp groups, routes them to Rides, Needs Review, Upcoming Jobs, Final Bid, Driver Recommendations, Schedules, Linked Rides, and Calendar workflow.

OTS Worker System

Scrapes ride cards from the OTS supplier portal, dedupes references, enriches routes with geocoding/OSRM, formats rows, appends to Master sheet, and archives expired jobs.

3. Unified Data Model

Both sources should feed one central schema so that the dashboard does not care whether the ride came from WhatsApp, OTS, or manual entry.

ENTITY	PURPOSE	IMPORTANT FIELDS
rides	Main ride record from all sources.	ride_id, source, pickup_date, pickup_time, pickup, drop_off, fare, vehicle_required, status
drivers	Driver profile and availability.	driver_id, name, phone, status, current_location, working_hours
vehicles	Independent vehicle resource.	vehicle_id, type, seats, registration, status
recommendations	AI/logic based dispatch suggestions.	ride_id, driver_id, vehicle_id, score, reason, status
driver_schedule	Driver timeline and availability.	driver_id, ride_id, start_time, end_time, next_available_time, current_location
vehicle_schedule	Vehicle booking protection.	vehicle_id, ride_id, driver_id, start_time, end_time, status
linked_rides	Return ride and chaining opportunities.	first_ride_id, second_ride_id, driver_id, vehicle_id, time_gap, saving_estimate
calendar_events	Calendar creation tracking.	ride_id, event_id, status, error_reason
archived_rides	Old completed ride history.	ride data snapshot, archived_at, archive_reason

4. Data Cleanup And Retention

The database should stay fast and clean. Old trash should not remain in active operational tables, but future rides must be protected.

RIDE TYPE	RETENTION RULE	ACTION
Future rides	Keep until ride date passes, even if the ride is 40+ days ahead.	Stay active
Assigned future rides	Keep active until completed or cancelled.	Stay active
Completed rides	Keep active for 10 days after completion.	Archive after 10 days
Cancelled, failed, duplicate, invalid rides	Keep briefly for audit.	Delete or archive after 3-5 days
Needs Review unresolved	Keep for operator review.	Archive after 7-10 days if unresolved
Expired linked opportunities	Close when timing is no longer valid.	Mark Expired

A daily cleanup worker should run automatically, for example at 2 AM. It should archive old completed rides, delete trash, expire old linked opportunities, and trim old logs.

5. Linked Ride Intelligence

Every new ride should be compared against upcoming and assigned rides to find return ride opportunities. This helps reduce empty mileage and improve driver productivity.

Matching Criteria

- Previous ride drop location is close to the next ride pickup location.
- Vehicle category is compatible.
- Time gap is realistic for the driver to complete both rides.
- Driver and vehicle are both available.
- Rest/fatigue rules are respected.
- Airport codes, postcodes, cities, and zones are normalized before matching.

Search Window

- Same day rides should get highest priority.
- Next 24 hours and next 3 days should be searched for near matches.
- Long future rides, such as a ride 40 days ahead, should remain searchable for linked opportunities around that ride date.
- If no exact match exists, the system should still show best-effort suggestions with lower confidence.

6. Dashboard Screens

SCREEN	PURPOSE
Dashboard	Today jobs, upcoming jobs, pending approvals, assignments, calendar failures, linked opportunities, revenue.
Live Rides	All incoming rides from WhatsApp and OTS with source filters.
Needs Review	Incomplete or low-confidence rides requiring operator correction.
Final Bid	Qualified rides ready for dispatch decision.
Driver Recommendations	Recommended driver/vehicle with score and reason. Operator approves here.
Linked Rides	Return ride opportunities, chaining score, saving estimate, and approval status.
Drivers	Driver status, current location, working hours, schedule, availability.
Vehicles	Vehicle type, seats, registration, current booking status.
Schedules	Driver and vehicle timeline views.
Calendar Monitor	Calendar created, failed, retry needed, and failure reason.
Reports	Completed jobs, revenue, driver utilization, linked ride savings.

7. Operator Workflow

1. New ride arrives from WhatsApp or OTS
2. System normalizes ride into central database
3. System routes ride to Live Rides, Needs Review, or Final Bid
4. Recommendation engine selects best driver and vehicle
5. Operator approves recommendation on dashboard
6. Assignment is created
7. Driver and vehicle schedule update automatically
8. Linked ride opportunities are shown
9. Calendar event is created or failure reason is shown
10. Old data is archived or deleted by cleanup worker

8. Future Add-On: Auto Bidding

A future production add-on can automatically bid on rides when the system criteria match. This should not replace operator control at first; it should start as an assisted mode and then move to full automation once confidence is proven.

Auto Bidding Criteria

- Fare is above configured minimum.
- Pickup/dropoff locations are within service area.
- Required vehicle type is available.
- Driver and vehicle schedule have no conflicts.
- Linked ride opportunity exists or pickup distance is acceptable.
- Rest/fatigue rules are satisfied.
- Route profitability score meets threshold.

Auto Bidding Dashboard Behavior

- System detects a matching ride from OTS or WhatsApp.
- System calculates dispatch score and bid score.
- If criteria match, system prepares or places the bid automatically depending on mode.
- Dashboard shows the bid as **Auto Bid Suggested** , **Auto Bid Placed** , or **Manual Approval Needed** .
- Operator can approve, reject, or override.
- All auto bidding decisions are stored in audit logs.

Recommended Rollout

MODE	DESCRIPTION	RISK LEVEL
Suggest Only	System shows that it would bid, but operator clicks approve.	Low
Assisted Auto Bid	System auto-bids only on high-confidence rides and shows result on dashboard.	Medium
Full Auto Bid	System bids automatically under strict rules and audit logging.	Higher, after validation

9. Recommended Production Architecture

Frontend

React or Next.js dashboard with operator actions, filters, live ride boards, and schedule views.

Backend API

Node.js API for rides, recommendations, assignment approval, calendar retry, reports, and settings.

Database

Supabase/Postgres as source of truth. Google Sheets can remain as reporting/export layer.

Workers

WhatsApp Worker, OTS Worker, Recommendation Worker, Calendar Worker, Cleanup Worker, and future Auto Bidding Worker.

10. Implementation Roadmap

PHASE	GOAL	OUTCOME
Phase 1	Dashboard MVP over current systems	Read Google Sheets/current pipeline and show unified operational view.
Phase 2	Central database	Move rides, schedules, recommendations, and linked rides into Postgres.
Phase 3	Operator actions	Approve recommendations, assign drivers, retry calendar, edit review rides.
Phase 4	Linked ride intelligence	Return ride matching, savings estimate, and chain assignment.
Phase 5	Auto cleanup	Archive/delete old data while protecting future rides.
Phase 6	Auto bidding add-on	Criteria-based bidding with dashboard visibility and audit logs.

11. Final Vision

The final system becomes a professional ACE Taxi Dispatch Control Panel. Operators will not need to jump between multiple sheets and scripts. WhatsApp and OTS rides will land in one place, the system will recommend the best driver and vehicle, linked ride opportunities will be highlighted, old data will be cleaned automatically, and calendar/bidding workflows will be visible from the dashboard.